

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Main)-2022

TEST DATE: 06-02-2022

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **-1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer. There is no negative marking.

Physics

PART – A

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

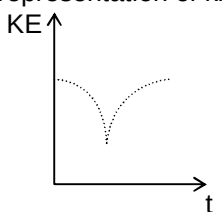
- If $\vec{E}(x, t) = E_0 \sin(\omega t - kx) \hat{j}$ and $\vec{B}(x, t) = B_0 \sin(\omega t - kx) \hat{k}$ are the electric field and magnetic field vectors respectively in an electromagnetic wave. What would be the direction of propagation of the EM Wave?

(A) $\hat{i}$
 (B) $\hat{i} + \hat{j}$
 (C) $\hat{k}$
 (D) $\hat{j} + \hat{k}$
- The percentage modulation in AM Wave is 60% and the peak value of information voltage is 18 Volt. Then the sum of maximum voltage $V_{\max}$ and minimum voltage $V_{\min}$ is

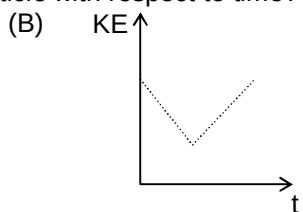
(A) 36 V
 (B) 60 V
 (C) 30 V
 (D) 72 V
- If the radiation of highest wavelength in the ultraviolet region of hydrogen spectrum is just able to eject photoelectrons from a metal. The value of threshold frequency for the given metal is:

(A) 3.83×10^{15} Hz
 (B) 4.33×10^{14} Hz
 (C) 2.46×10^{15} Hz
 (D) 7.83×10^{14} Hz
- A particle is projected with speed u at an angle θ with horizontal. Which of the following is the correct representation of kinetic energy of the particle with respect to time? (Take $g = 10 \text{ m/s}^2$)

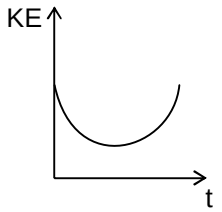
(A)



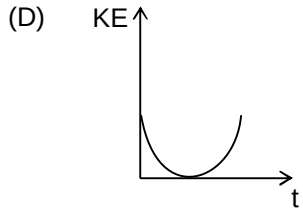
(B)



(C)

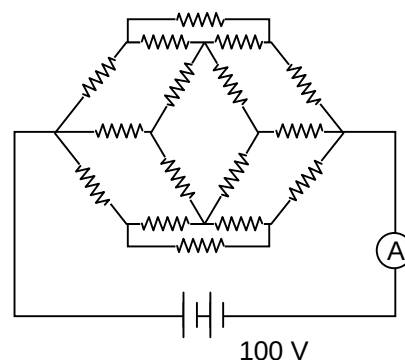


(D)

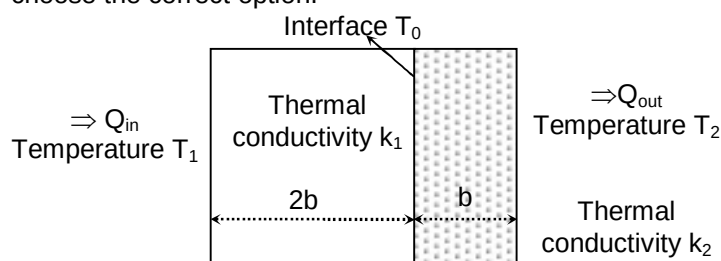


5. In the given DC circuit diagram each resistor R is of $65\ \Omega$. Find the current through ammeter?

- (A) $\frac{5}{3}\text{ A}$
 (B) $\frac{3}{5}\text{ A}$
 (C) $\frac{7}{3}\text{ A}$
 (D) $\frac{3}{7}\text{ A}$

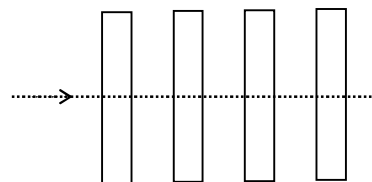


6. In a composite slab as shown, the temperature at the interface between two materials is equal to the average of the temperature at the two ends. Assuming steady heat flow through conduction, choose the correct option.



- (A) $2k_1 = k_2$
 (B) $k_1 = k_2$
 (C) $2k_1 = 3k_2$
 (D) $k_1 = 2k_2$

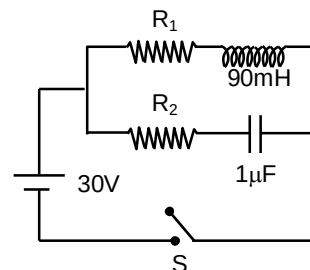
7. A beam of unpolarized light of intensity I_0 falls on a system of four identical linear polarizers placed in a line as shown in the figure. The transmission axis of any two successive polarizers make an angle of 30° with each other. If the transmitted light has intensity I , the ratio $\frac{I}{I_0}$ is



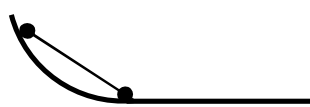
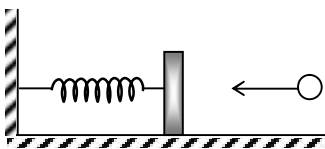
- (A) $\frac{81}{256}$
 (B) $\frac{9}{16}$
 (C) $\frac{27}{64}$
 (D) $\frac{27}{128}$

8. If in the circuit shown, the current through the ideal battery remains constant after the switch S is closed. Then the values of R_1 and R_2 respectively are (an ideal inductor and capacitor are used)

- (A) $300\ \Omega$ each
 (B) $200\ \Omega$ and $300\ \Omega$
 (C) $300\ \Omega$ and $200\ \Omega$
 (D) $200\ \Omega$ each

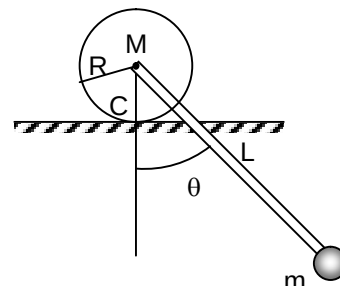


9. Two smooth spheres of radii R and $2R$, specific gravity 0.1 and 0.2 respectively are released from the bottom of a sufficiently deep-water tank (coefficient of viscosity 10 poise). The spheres travel all the way to the surface of tank, emerges out and goes up to height H_1 and H_2 respectively in air. Then,
- (A) $H_1 = H_2$
 (B) $H_1 > H_2$
 (C) $H_1 < H_2$
 (D) can't be determined
10. Arrange the following telescopes, in order of decreasing resolving power: (where D is the telescope diameter and λ is the wavelength)
- (I) $D = 100 \text{ m}$, $\lambda = 21 \text{ cm}$
 (II) $D = 2 \text{ m}$, $\lambda = 500 \text{ nm}$
 (III) $D = 1 \text{ m}$, $\lambda = 100 \text{ nm}$
 (IV) $D = 2 \text{ m}$, $\lambda = 10 \text{ mm}$
- (A) (III), (II), (IV), (I)
 (B) (II), (III), (I), (IV)
 (C) (IV), (III), (II), (I)
 (D) (III), (II), (I), (IV)
11. For a gas under isothermal conditions its pressure P varies with volume V as $P \propto V^{-2/3}$. Then the bulk modulus B will be proportional to
- (A) $V^{-2/3}$
 (B) $V^{-5/3}$
 (C) V^{-2}
 (D) $V^{-\frac{2}{5}}$
12. When a liquid is placed in vacuum (e.g. a leak from a space vehicle) it evaporates very fast and can lead to rapid cooling. A water droplet at 10°C escapes into space and begins to evaporate. Find its temperature when it has lost 1% of its mass. (To heat 1 kg of water by 1°C requires 4.2 kJ of energy; to evaporate 1 kg of water requires 2.26 MJ of energy)
- (A) 5.6°C
 (B) 4.6°C
 (C) 4.2°C
 (D) 5.4°C
13. Three infinitely long wires, each carrying equal current are placed in the xy -plane along $x = 0$, $+d$ and $-d$. On the xy -plane, the magnetic field vanishes at
- (A) $x = \pm \frac{d}{2}$
 (B) $x = \pm d \left(1 + \frac{1}{\sqrt{3}} \right)$
 (C) $x = \pm d \left(1 - \frac{1}{\sqrt{3}} \right)$
 (D) $x = \pm \frac{d}{\sqrt{3}}$

14. The velocity $v(x)$ of a particle moving in one dimension is given by $v(x) = v_0 \sin\left(\frac{\pi x}{x_0}\right)$, where v_0 and x_0 are positive constants of appropriate dimensions. If the particle is initially at $\frac{x}{x_0} = \varepsilon$, where $|\varepsilon| \ll 1$, then in the long time, it
- Executes an oscillatory motion around $x = 0$
 - Tends towards $x = 0$
 - tends towards $x = x_0$
 - executes an oscillatory motion around $x = x_0$
15. Two electric dipoles P_1 and P_2 are placed at $(0, 0, 0)$ and $(1, 0, 0)$ respectively with both of them pointing in the $+z$ direction. Without changing the orientations of the dipoles P_1 is moved to $(0, 2, 0)$. The ratio of the electrostatic potential energy of the dipoles after moving to that before moving is
- $\frac{1}{16}$
 - $\frac{1}{2}$
 - $\frac{1}{4}$
 - $\frac{1}{8}$
16. A light rigid rod with two equal masses m at the ends, is held with one end on a horizontal surface. The other end rests on a circular curve of radius 3.4 m, at a distance from the horizontal corresponding to $1/8$ th of the circumference of a circle, as shown in Figure. If the surface is frictionless, what is the speed of the rod and masses when released and both masses slide on the flat surface? The masses remain in the same vertical plane.
- 
- 2.1 m/s
 - 3.1 m/s
 - 4.1 m/s
 - 5.1 m/s
17. A block of mass 0.38 kg is kept at rest on a frictionless surface and attached to a wall with a spring of negligible mass. A bullet weighing 0.02 kg moving with a speed of 200 m/s hits the block at time $t = 0$ and gets stuck to it. The equation of displacement of the block (in meter) with respect to the equilibrium position is given by (spring constant = 640 N/m)
- 
- $-2 \sin(5t)$
 - $-\cos(10t)$
 - $-0.4 \cos(25t)$
 - $-0.25 \sin(40t)$
18. A spherical capacitor consists of two concentric conducting spheres of radii a and b ($a > b$). The outer sphere is grounded and a charge Q is placed on the inner sphere. The outer conductor then contracts from radius a to radius a' . Find the work done by the electric force.
- $\frac{Q^2(a - a')}{8\pi\epsilon_0 aa'}$

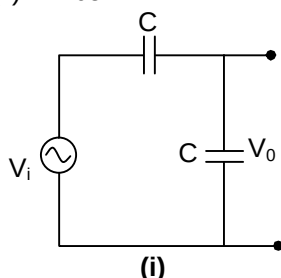
- (B) $\frac{Q^2(a-a')}{6\pi\epsilon_0 aa'}$
- (C) $\frac{Q^2(a-a')}{4\pi\epsilon_0 aa'}$
- (D) $\frac{Q^2(a-a')}{2\pi\epsilon_0 aa'}$

19. A mass m is attached to one end of a weightless stiff rod of length L which is rigidly connected to the center of a uniform cylinder of radius R as shown in the figure. Assuming that the cylinder rolls without slipping, the natural angular frequency of oscillation of the system will be (for small oscillation)?

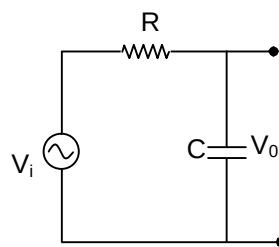


- (A) $\sqrt{\frac{2mgL}{\frac{3MR^2}{2} + m(L-R)^2}} \text{ rad/s}$
- (B) $\sqrt{\frac{2mgL}{\frac{MR^2}{2} + m(L-R)^2}} \text{ rad/s}$
- (C) $\sqrt{\frac{mgL}{\frac{3MR^2}{2} + m(L-R)^2}} \text{ rad/s}$
- (D) $\sqrt{\frac{mgL}{MR^2 + m(L-R)^2}} \text{ rad/s}$

20. The phase difference (δ) between input V_i and output voltage V_o for the following circuit (i) and (ii) will be



(i)



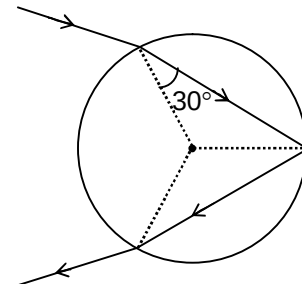
(ii)

- (A) 0 and 0
- (B) $\pi/2$ and $0 < \delta < \frac{\pi}{2}$ respectively
- (C) $\frac{\pi}{2}$ and $\frac{\pi}{2}$
- (D) 0 and $0 < \delta < \frac{\pi}{2}$ respectively

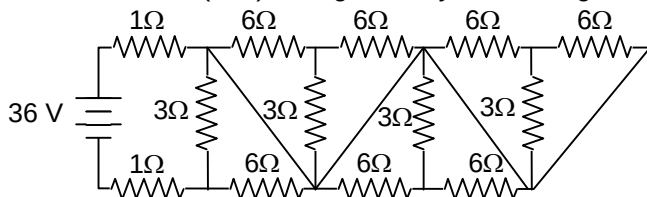
SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. A ray incident on a glass sphere of refractive index $\sqrt{2}$ as shown in figure. Angle of refraction of the ray inside sphere is 30° . If total deviation suffered by the ray is $30x^\circ$, then find x .

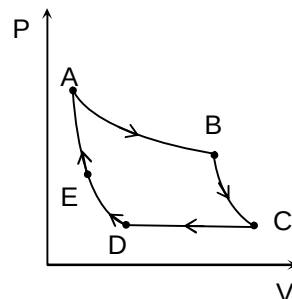


22. Find the current (in A) through battery in following circuit.



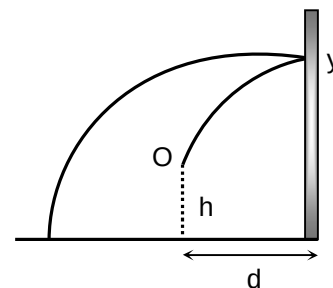
23. The voltage on a capacitor in a certain circuit is given by $V(t) = V_0 e^{-t/RC}$. If $R = 50 \text{ k}\Omega \pm 5\%$ and $C = 0.01 \text{ }\mu\text{F} \pm 10\%$ and the fractional error in the voltage at $t = 50 \text{ }\mu\text{s}$ is N . Find the value of $100N$.
24. An alternating current is expressed as $i = 2\cos\omega t + \sqrt{2}\sin\omega t$. The RMS value of current is N . Find the value of N ?

25. An ideal gas undergoes an isothermal expansion along a path AB, adiabatic expansion along BC, isobaric compression along CD, isothermal compression along DE, an adiabatic compression along EA, as shown in the figure. The work done by the gas along the process BC is 10J. The change in the internal energy along process EA is 16J. The absolute value of the change in the internal energy (in Joule) along the process CD is N . Find the value of N ?

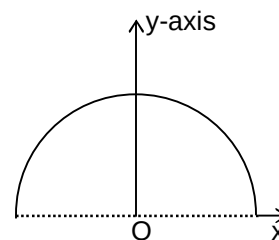


26. Two transverse waves $y_1 = 5 \cos(kx - \omega t)$ cm, and $y_2 = 5 \cos(kx + \omega t)$ cm, travel on a string along x -axis. If the speed of a point at $x = 0$ is zero at $t = 0 \text{ s}$, 0.25 s and 0.5 s , then the minimum frequency (in Hz) of the waves is
27. Consider N_1 number of ideal gas particles enclosed in a volume V_1 . If the volume is changed to V_2 and the number of particles is reduced by half then the mean free path becomes four times of its initial value. The value of $\frac{V_1}{V_2}$ is X . Find the value of X ?

28. A ball is thrown from a height $h = 0.5\text{m}$ above the ground. The ball leaves the point located at distance $d = 10\text{m}$ from the wall at 45° to the horizontal with velocity $u = 20\text{m/s}$. How far from the wall (in horizontal direction) does the ball hit the ground, assuming collision from the wall to be elastic? (Take $g = 10\text{m/s}^2$)



29. A semicircular disc of radius $R = \pi$ meter and mass M has density ρ which varies as $\rho = Cr^2$, where r is the distance from the center of the base and C is a constant. The distance of center of mass of the semicircular disk from O is



30. A body slides down a rough plane inclined to the horizontal at 30° . If 70% of the initial potential energy is dissipated during the descent, then the coefficient of sliding friction is N . Find the value of $10N$? (Take root 3 as 1.73)

Chemistry

PART – B

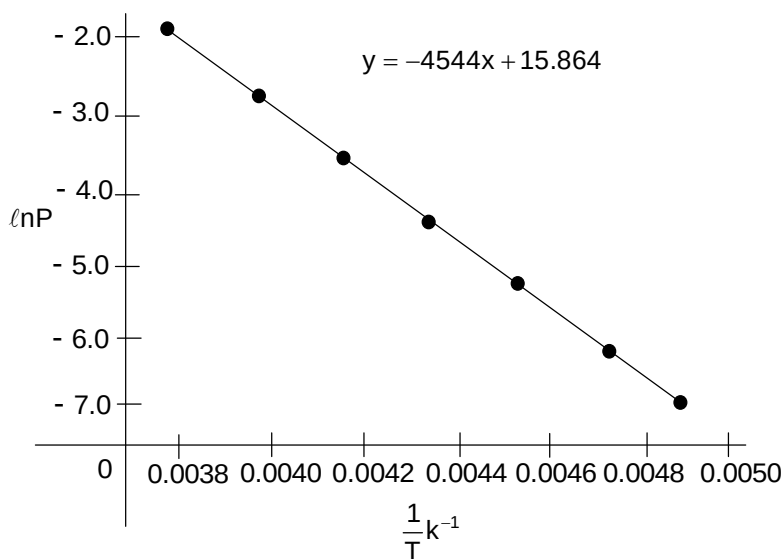
SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Permanganate ion oxidizes H_2O_2 in acidic solution according to the following equation

$$2\text{MnO}_4^- (\text{aq}) + 5\text{H}_2\text{O}_2 (\text{aq}) + 6\text{H}^+ (\text{aq}) \longrightarrow 2\text{Mn}^{2+} (\text{aq}) + 5\text{O}_2 (\text{g}) + 8\text{H}_2\text{O} (\ell)$$

 If 35.0 ml of an acidic 0.150 M KMnO_4 solution is required to consume all the H_2O_2 in 50.0 ml of a disinfectant solution, what is the concentration of H_2O_2 in the disinfectant?
 (A) 0.042 M
 (B) 0.105 M
 (C) 0.263 M
 (D) 0.368 M
32. Decomposition of 1 gm of which of the following compounds into its constituent elements gives the greatest amount of N_2 gas?
 (A) NO
 (B) NO_2
 (C) N_2O_4
 (D) NH_3
33. The natural logarithm of the vapour pressure (in bar) of HCN (s) as a function of the reciprocal of the absolute temperature is plotted below:

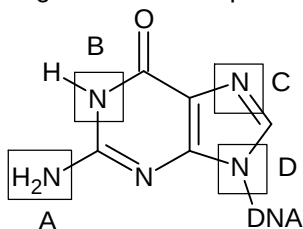


What is heat of sublimation of HCN?

- (A) 4.54 kJ/mole
 (B) 15.9 kJ/mole
 (C) 37.8 kJ/mole
 (D) 132 kJ/mole

34. Which changes to this galvanic cell increase the measured potential?
 $\text{Cu(s)} / \text{Cu}^{2+}(\text{aq}), 0.10\text{M} \parallel \text{Ag}^{+}(\text{aq}), 1.0\text{M} / \text{Ag(s)}$
- I. Increasing $[\text{Cu}^{2+}]$ in the $\text{Cu} / \text{Cu}^{2+}$ half-cell to 0.5 M.
- II. Adding Cl^{-} to the $\text{Ag}^{+} / \text{Ag}$ half-cell until $[\text{Cl}^{-}] = 0.01\text{M}$.
- (A) Only I
 (B) Only II
 (C) Both I and II
 (D) Neither I nor II
35. Rank the element Si, P, Ge, and As in increasing order of their first ionization energy
- (A) $\text{Si} < \text{P} < \text{Ge} < \text{As}$
 (B) $\text{As} < \text{Ge} < \text{P} < \text{Si}$
 (C) $\text{Ge} < \text{Si} < \text{As} < \text{P}$
 (D) $\text{Ge} < \text{As} < \text{Si} < \text{P}$
36. The isoelectric point of a protein is the pH at which it is electrically neutral. Which mutation of an amino acid $\text{NH}_2 - \underset{\text{R}}{\text{CH}} - \text{COOH}$ in the protein would have the greatest effect on its isoelectric point, assuming that the mutation does not significantly affect the protein's overall structure?
- (A) Serine ($\text{R} = \text{CH}_2 - \text{OH}$) $\longrightarrow$ Lysine ($\text{R} = \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{NH}_2$)
 (B) Glutamine ($\text{R} = \text{CH}_2 - \text{CH}_2 - \text{CONH}_2$) $\longrightarrow$ Methionine ($\text{R} = \text{CH}_2 - \text{CH}_2 - \text{S} - \text{CH}_3$)
 (C) Isoleucine ($\text{R} = \text{CH}(\text{CH}_3)\text{CH}_2 - \text{CH}_3$) $\longrightarrow$ Valine $\left(\text{R} = \text{HC} \begin{array}{l} \nearrow \text{CH}_3 \\ \searrow \text{CH}_3 \end{array} \right)$
 (D) Alanine ($\text{R} = \text{CH}_3$) $\longrightarrow$ Glycine ($\text{R} = \text{H}$)
37. The melting point of silicon dioxide is (1713°C) is higher than the melting point of silicon (1414°C). What is the best explanation for this difference?
- (A) Silicon – oxygen bond are stronger than Silicon – Silicon bonds.
 (B) Silicon – dioxide is an ionic solid while Silicon is metallic solid.
 (C) Silicon – dioxide is polar while Silicon is non-polar.
 (D) Silicon – dioxide forms tetragonal crystal while Silicon forms cubic crystals.
38. Metallic gold will not dissolve in nitric acid alone, but will dissolve in aqua regia. Which is the best explanation for this observation?
- (A) Chloride ion complexes gold (III), stabilizing it as AuCl_4^{-} .
 (B) Nitric acid is a stronger oxidant in more acidic solution.
 (C) Nitric acid and hydrochloric acid react to form NO_2Cl , which react readily with metallic gold.
 (D) Nitric and hydrochloric acid react to form Cl_2 , which react readily with metallic gold.

39. Guanine is one of the bases in DNA, is susceptible to attack by alkylating agents. Which nitrogen in guanine is most prone to alkylation?

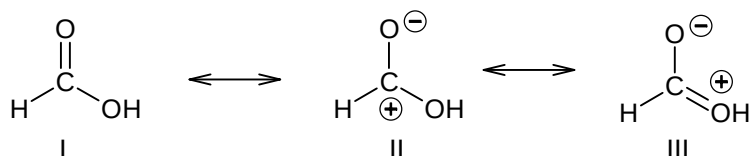


- (A) A
(B) B
(C) C
(D) D
40. Sodium halide crystallize in the rock salt structure, which has a cubic unit cell with 4 formula units per unit cell. If sodium halide has density of 2.83 g/cm^3 and a unit cell edge length of 462 pm. Which compound is it?
(A) NaF
(B) NaCl
(C) NaBr
(D) NaI
41. The degree of hydrolysis of a salt of weak acid and weak base in its 0.1 M solution is found to be 50%. If the molarity of the solution is 0.2 M, the percentage hydrolysis of the salt should be:
(A) 100%
(B) 50%
(C) 25%
(D) None of these
42. Solution of chemical compound x react with AgNO_3 solution to form white ppt. 'Y', which dissolves in NH_4OH to give a compound 'Z'. When 'Z' is treated with dil. HNO_3 , 'Y' reappears. The chemical compound 'X' can be
(A) NaCl
(B) CH_3Cl
(C) NaBr
(D) NaI
43. The $(n_2 + n_1)$ and $(n_2^2 - n_1^2)$ for He^+ ion in atomic spectrum are 4 and 8 respectively. The wave length of emitted photon when electron jump from n_2 to n_1 is
(A) $\frac{32}{9}R_H$
(B) $\frac{9}{32}R_H$
(C) $\frac{9}{32R_H}$
(D) $\frac{32}{9R_H}$

44. The vapour pressure of a saturated solution of sparingly soluble salt (XCl_3) was 17.20 mm of Hg at 27°C . If the vapour pressure of pure H_2O is 17.25 mm of Hg at 300 K, what is the solubility of sparingly soluble salt XCl_3 in mole/lit?

- (A) 4.04×10^{-2}
 (B) 4.08×10^{-2}
 (C) 2.02×10^{-2}
 (D) 4.04×10^{-3}

45.



Among these canonical structure, the correct order of stability is

- (A) $\text{I} > \text{II} > \text{III}$
 (B) $\text{III} > \text{II} > \text{I}$
 (C) $\text{I} > \text{III} > \text{II}$
 (D) $\text{II} > \text{I} > \text{III}$
46. Sulphuric acid cannot be used as:
 (A) Hygroscopic agent
 (B) Oxidizing agent
 (C) Sulphonating agent
 (D) Efflorescent
47. n moles of 'He' gas are placed in a vessel of volume V lit. At TK, if V_1 is free volume of 'He' then diameter of 'He' atom is
- (A) $\left[\frac{3 V_1}{2 \pi N_A n} \right]^{1/3}$
 (B) $\left[\frac{3(V - V_1)}{2 \pi N_A n} \right]^{1/3}$
 (C) $\left[\frac{6(V - V_1)}{2 \pi N_A n} \right]^{1/3}$
 (D) $\left[\frac{3 V_1}{\pi N_A n} \right]^{1/3}$
48. Blister copper is refined by stirring molten impure metal with green logs of wood because such a wood liberates hydrocarbon gases. This process X is called _____ and the metal contains impurities of Y is _____
- (A) X = Cupellation; Y = CuO_2
 (B) X = Polling; Y = Cu_2O
 (C) X = Polling; Y = CuO
 (D) X = Cupellation; Y = CuO

49. 0.001 mole of $\text{Co}(\text{NH}_3)_5(\text{NO}_3)(\text{SO}_4)$ was passed through a cation exchanger and the acid coming out of it required 20 ml of 0.1 M NaOH for neutralization. Hence, the complex is:
- (A) $[\text{Co}(\text{SO}_4)(\text{NH}_3)_5]\text{NO}_3$
 (B) $[\text{Co}(\text{NO}_3)(\text{NH}_3)_5]\text{SO}_4$
 (C) $[\text{Co}(\text{NH}_3)_5]\text{SO}_4\text{NO}_3$
 (D) None of these
50. First generation automobile air bags were inflated by decomposition of Barium azide. What mass of barium azide would be required to inflate a 20 L air bag to a pressure of 1.23 atm at 27°C ? (Given at. Mass of Ba = 137.327 and N = 14).
- (A) 78.6 gm
 (B) 74 gm
 (C) 157 gm
 (D) 52.4 gm

SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXX.XX).

51. The specific heat capacity of iron is $0.461 \text{ J g}^{-1} \text{ K}^{-1}$ and that of titanium is $0.544 \text{ J g}^{-1} \text{ K}^{-1}$. A sample consisting of 10.0 g Fe and 10 g of titanium at 100.0°C losses 200 J of heat to the environment. What is the final temperature of the sample?
52. The K_{sp} of $\text{Ca}(\text{OH})_2$ at 298 K is $x \times 10^{-6}$ at 298 K
- | Half reaction | $E^\circ \text{V}$ |
|---|--------------------|
| $\text{Ca}^{2+}(\text{aq}) + 2e^- \longrightarrow \text{Ca}(\text{s})$ | -2.87 V |
| $\text{Ca}(\text{OH})_2(\text{s}) + 2e^- \longrightarrow \text{Ca}(\text{s}) + 2\text{OH}^-(\text{aq})$ | -3.02 V |
- Find x? [Given $\log 0.9263 = 8.44$]
53. Sulphur trioxide is formed from the reaction of sulphur dioxide and oxygen
- $$\text{SO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g})$$
- At 1000 K, an equilibrium mixture has partial pressure of 0.562 bar SO_2 , 0.101 bar O_2 and 0.332 bar SO_3 . What is the equilibrium constant K_p for this reaction at this temperature.
54. In an amino acid, the carboxylic group ionizes at $\text{pK}_{a_1} = 2.34$ and ammonium ion at $\text{pK}_{a_2} = 9.60$. The isoelectric point of the amino acid is at pH?
55. If the following molecule undergo dimerization then find the value of $\frac{yz}{x}$.
- $\text{ClO}_3, \text{OF}, \text{GaH}_3, \text{AlCl}_3, \text{ICl}_3, \text{BeH}_2, \text{NO}_2$
- Where x = Number of molecules which are hypervalent in dimeric form.
 y = Number of molecule which complete octet in dimer form.
 z = number of molecule which are hypovalent in dimeric form.

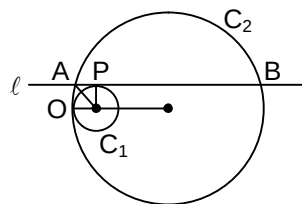
56. Find the number of basic radicals among the following cations, which can form soluble complex on adding excess of NH_3 .
 $\text{Cd}^{2+}(\text{aq}), \text{Pb}^{2+}(\text{aq}), \text{Ni}^{2+}(\text{aq}), \text{Mn}^{2+}(\text{aq}), \text{Zn}^{2+}(\text{aq}), \text{Ag}^+(\text{aq}), \text{Hg}^{2+}(\text{aq}), \text{Fe}^{3+}(\text{aq}), \text{Mg}^{2+}(\text{aq})$
57. Find the total number of electron of Co^{2+} ion which are present in non axial-d-orbitals in the complex compound $[\text{Co}(\text{dmg})_2]$
58. Calculate the I – I distance in $\overset{\circ}{\text{A}}$ for given compound H_2CCl_2 if C – I bond length is $2.35 \overset{\circ}{\text{A}}$
 $(\sin 60^\circ = 0.866)$
59. At constant temperature and pressure, 5.0 L of SO_2 is combined with 3.0 L of O_2 according to the equation
 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{SO}_3(\text{g})$
 After SO_3 formation is complete, what is the volume of the mixture?
60. An element 'E' has electronic configuration of 'K' shell is $(n-5)s^2$ and it has total number of electron in its outer most shell is 2, in penultimate shell is 8 and in anti penultimate shell is 25. Find the total number of unpaired electron in element 'E' in their ground state.

Mathematics**PART – C****SECTION – A****(One Options Correct Type)**

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Many states use a sequence of three letters followed by a sequence of three digits as their standard license-plate pattern. Given that each three-letter three-digit arrangement is equally likely, the probability that such a license plate will contain at least one palindrome (a three-letter arrangement or a three-digit arrangement that reads the same left-to-right as it does right-to-left) is $\frac{m}{n}$, where m and n are relatively prime positive integers, then the value of $m + n$
- (A) 58
(B) 59
(C) 60
(D) 63
62. There are 20 students participating in an after-school program offering classes in yoga, bridge, and painting. Each student must take at least one of these three classes, but may take two or all three. There are 10 students taking yoga, 13 taking bridge, and 9 taking painting. There are 9 students taking at least two classes, then number of students who are taking all three classes?
- (A) 1
(B) 2
(C) 3
(D) 4
63. Right triangle ABC has right angle at C and $\angle BAC = \theta$; the point D is chosen on AB so that $|AC| = |AD| = 1$; the point E is chosen on BC so that $\angle CDE = \theta$. The perpendicular to BC at E meets AB at F, then $\lim_{\theta \rightarrow 0} |EF|$ is equal to
- (A) $\frac{1}{2}$
(B) $\frac{1}{3}$
(C) $\frac{1}{4}$
(D) $\frac{1}{5}$
64. All values of the real parameter 'a' such that the nonnegative solutions of the equation $(2a - 1) \sin x + (2 - a) \sin 2x = \sin 3x$ form an infinite arithmetic progression are
- (A) $a < -2$ or $a > 2 \cup \{0\}$
(B) $a < -3$ or $a > 3 \cup \{0\}$
(C) $a < -5$ or $a > 5 \cup \{0\}$
(D) $a < -\frac{13}{2}$ or $a > \frac{13}{2} \cup \{0\}$

65. If $z = e^{i\pi/13}$, then $\frac{1}{1-z}$ is equal to (where $i = \sqrt{-1}$)
- (A) $1 - z^2 + z^4 - z^6 + z^8 - z^{10} + z^{12}$
 (B) $1 + z^2 + z^4 + z^6 + z^8 + z^{10} + z^{12}$
 (C) $z - z^3 + z^5 - z^7 + z^9 - z^{11}$
 (D) $z + z^3 + z^5 + z^7 + z^9 + z^{11}$
66. If $|z_1| = 1$, $|z_2 - 2| = 3$ and $|z_3 - 5| = 6$, then maximum value of $|2z_1 - 3z_2 - 4z_3|$ is (where z_1, z_2 and z_3 are complex numbers)
- (A) 43
 (B) 44
 (C) 45
 (D) 61
67. Let $P(x)$ be a polynomial of degree 4 having a local maximum at $x = 2$ and $\lim_{x \rightarrow 0} \left(3 - \frac{P(x)}{x} \right) = 27$. If $P(1) = -9$ and $P''(x)$ has a local minimum at $x = 2$. If global maximum value of $y = P'(x)$ on the set $A = \{x : x^2 + 12 \leq 7x\}$ is $4M$, then the value of M is
- (A) 6
 (B) 7
 (C) 9
 (D) 13
68. The parabola $y^2 = x - y$ divides the triangle formed by the tangent to it at the origin and the straight line $x + y = 0$ and $y = -1$ into two parts whose areas are A_1 and A_2 . If $\frac{A_1}{A_2} = \frac{p}{q}$ where $p, q \in \mathbb{N}$ and $\frac{p}{q}$ is in simplest form, then the value of $(p + q)$
- (A) 3
 (B) 4
 (C) 5
 (D) 6
69. Circles C_1 and C_2 of radii r and R respectively, touch each other as shown in the figure. The line ℓ , which is parallel to the line joining the centres of C_1 and C_2 is tangent to C_1 at P and intersects C_2 at A, B . If $R^2 = 2r^2$, then $\angle AOB$ equals
- (A) $22\frac{1}{2}^\circ$
 (B) 45°
 (C) 60°
 (D) $67\frac{1}{2}^\circ$
70. Let $x > 0$ be a fixed real number. Then the integral $\int_0^\infty e^{-t} |x - t| dt$ is equal to
- (A) $x + 2e^{-x} - 1$
 (B) $x - 2e^{-x} + 1$
 (C) $x + 2e^{-x} + 1$
 (D) $-x - 2e^{-x} + 1$



71. The largest perfect square that divides $2014^3 - 2013^3 + 2012^3 - 2011^3 + \dots + 2^3 - 1^3$ is
 (A) 1^2
 (B) 2^2
 (C) 1007^2
 (D) 2014^2
72. The sum of non-real roots of the polynomial equation $x^3 + 3x^2 + 3x + 3 = 0$
 (A) equals 0
 (B) lies between 0 and 1
 (C) lies between -1 and 0
 (D) has absolute values bigger than 1
73. If a, b are positive real numbers such that the lines $ax + 9y = 5$ and $4x + by = 3$ are parallel, then the least possible value of $a + b$ is
 (A) 13
 (B) 12
 (C) 8
 (D) 6
74. For real x with $-10 \leq x \leq 10$ define $f(x) = \int_{-10}^x 2^{[t]} dt$ (where $[.]$ denotes greatest integer function).
 The number of points of discontinuity of $f(x)$ in the interval $(-10, 10)$ is
 (A) 0
 (B) 10
 (C) 18
 (D) 19
75. If $(1-x^3)^n = \sum_{r=0}^n a_r x^r (1-x)^{3n-2r}$, then a_r is (where $n \in \mathbb{N}$ natural number)
 (A) ${}^nC_r 3^r$
 (B) ${}^nC_r \left(\frac{1}{3}\right)^r$
 (C) ${}^nC_{2r} 3^r$
 (D) None of these
76. Number of solution of the equation $\sin^{-1}(\sin x) = x - \pi$; $x \in \left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$ is
 (A) 0
 (B) 1
 (C) 2
 (D) 3
77. Let $S = \{1, 2, 3, 4, 5, 6, 7\}$ P is a subset of S. After replacing back in S the elements of P, again subset Q is selected. The number of ways such that $n(P \cap Q) = 2$, $n(P \cup Q) = 4$ is
 (A) 810
 (B) 820
 (C) 830
 (D) 840

78. The general solution of $\frac{dy}{dx} = \frac{4x - 5y + 3}{5x + 6y - 2}$ is
- (A) $2x^2 - 5xy - 3y^2 + 3x + 2y + c = 0$
 (B) $2x^2 + 5xy + 3y^2 + 3x + 2y = c$
 (C) $3x^2 - 5xy + 2y^2 + 2x + 3y = c$
 (D) $4x^2 - 5xy + 6y^2 + 3x - 2y = c$
79. "If you are born in India then you are citizen of India" contrapositive of this statement is
- (A) if you are born in India then you are not citizen of India
 (B) if you are not citizen of India then you are not born in India
 (C) if you are citizen of India then you are not born in India
 (D) if you are citizen of India then you are born in India
80. The angle between the lines whose direction cosines are given by $l + m + n = 0$ and $2l^2 + 2m^2 - n^2 = 0$
- (A) $\pi - \cos^{-1}\left(\frac{1}{3}\right)$
 (B) $\pi - \cos^{-1}\left(\frac{1}{4}\right)$
 (C) $\cos^{-1}\left(\frac{1}{4}\right)$
 (D) $\frac{\pi}{3}$

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81. The minimum value of $\frac{9x^2 \sin^2 x + 4}{x \sin x}$ for $0 < x < \pi$
82. A triangle has vertices $P = (-8, 5)$, $Q = (-15, -19)$ and $R = (1, -7)$. The equation of the bisector of $\angle P$ can be written in the form $ax + 2y + c = 0$, then the value of $a + c$
83. Let a, b, c be the three sides of a triangle, and let α, β, γ be the angles opposite them. If $a^2 + b^2 = 1989c^2$, then the value of $\frac{\cot \gamma}{\cot \alpha + \cot \beta}$ is equal to
84. Let $f(x) = |x - p| + |x - 15| + |x - p - 15|$, where $0 < p < 15$. Then the minimum value taken by $f(x)$ for x in the interval $p \leq x \leq 15$ is equal to
85. For any integer $n \geq 5$, let there are two $n \times n$ invertible matrices with real entries A, B satisfy the equation $A^{-1} + B^{-1} = (A + B)^{-1}$. If $|A| = 3$, then the value of $|B|$
86. The number of 2×2 matrices A with elements from the set $\{-1, 0, 1\}$ such that $A^2 = I$ is
87. Shortest distance between the lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$ is

88. Number of common tangents with finite slope to the curve $(x - 1)(y + 2) = 9$ and $(y + 2)^2 = 8(x - 1)$ is
89. Let $\vec{r}$ be a variable vector and $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ such that the scalars $(\vec{r} \cdot \hat{i}), (\vec{r} \cdot \hat{j}), (\vec{r} \cdot \hat{k})$ are positive integers. If $\vec{r} \cdot \vec{a}$ is not greater than 10, then total number of possible vectors $\vec{r}$ is given by
90. A student obtained the mean and standard deviation of 100 observations as 40 and 5.1 respectively. It was later found that he had wrongly copied down an observation as 50 instead of 40. The correct standard deviation is